DTU Compute
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Machine Learning for Semi-Automated Annotations of the Linear A Inscriptions

Bachelor Thesis

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Preface

This Bachelor's thesis was prepared at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark in fulfillment of the requirements for acquiring a Bachelor of Science in Engineering (BSc Eng) degree in General Engineering (Cyber Systems).

Kongens Lyngby, March 4, 2026

Frederik W. L. Christoffersen

Abstract

This thesis contains a thorough description of the problem to be solved, including the context necessary to fathom the importance of solving it. It also outlines an initial miniature literature review that provides an overview of the most relevant work and sources to be further studied and cited. Finally, it covers the associated plan and timeline that will guide the project's execution, including a Gantt chart and key milestones.

Acknowledgements

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Co-supervisor

Generative AI tools were occasionally used as support for brainstorming, linguistic editing, code exploration, and research during the development of this project.

Contents

Acronyms

SAM Segment Anything Model

Cefael Collections de l'École française d'Athènes en ligne

PDR Project Definition Report

IPR Intellectual Property Rights

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CHAPTER 1
Introduction

1.1 Background

When the British archaeologist Arthur Evans in the beginning of the 20th century went to excavate Knossos on Crete, he found a huge amount of clay tablets written in a previously unknown linear script. Some documents were clearly of a newer script, which he called *Linear B*, while others were of an older one, which he named *Linear A*. Some fifty years later, the British amateur archaeologist Michael Ventris finally succeeded in deciphering the newer script Linear B, which he proved to be representing Mycenaean Greek, the earliest attested form of Greek. But the same hypothesis did not hold for the older script Linear A, whose encoded language remained unknown, still yet to be deciphered.

A key obstacle to its decipherment has been the small size of its corpus, lately consisting of only around 2,500 inscriptions (about 1,400 if only including those well-preserved enough to be readable) - significantly smaller than that of its younger counterpart that now spans more than 4,500 inscriptions (which are comparatively longer and better preserved).[1,2] But new inscriptions are still being discovered, with the corpus most recently extended in 2025 by over 100 new documents.[3,4]

Another challenge has been the lack of digital representations of these inscriptions, in a form suitable for statistical analysis. While photographs with accompanying drawings of nearly the entire corpus are available online through Collections de l'École française d'Athènes en ligne (Cefael) (see Godart and Olivier (1976–1985)), these are only available in print

[1] Salgarella 2020.
[2] Salgarella 2025, p. 13, 46.
[3] Del Freo and Zurbach 2025.
[4] Consiglio Nazionale delle Ricerche 2025.

form. This motivated the development of the palæographical[5] database *The Signs of Linear A* (SigLA), which seeks to ”[develop] a systematic, exhaustive and user-friendly open access database of all Linear A inscriptions [...] in order to carry out statistical and palæographic analyses within the epigraphic corpus” (Salgarella and Castellan 2020). However, the last addition to SigLA was in 2021, and the project has so far managed to document just 772 of all currently known inscriptions.[6,7]

1.1.1 A problem of manual digitization

The drawings currently stored on SigLA were made by Ester Salgarella using the painting tool Krita. For each document, she manually created her own annotated drawings based on the originals from Cefael. From these, she proceeded to create layered Krita files with metadata for each sign, which could then at last be imported to SigLA.[8] According to her (personal communication), this has been a highly time-consuming undertaking. And the necessity of this kind of labor-intensive work has arguably been the biggest reason to why SigLA still does not contain the entire available corpus.

The aim of this project is therefore to investigate whether parts of this process can be automated. More specifically, whether some kind of semi-automatic segmentation tool can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database.

1.1.2 A problem of small data

The small size of the Linear A corpus and the irregularities in quality of its documents is a huge problem for the application of deep learning approaches, which typically require large amounts of data to perform well. Consequently, this project will instead seek to make use of generalized

[5] The term ”palæographical” refers to the study of ancient writing systems.
[6] SigLA 2021a.
[7] SigLA 2021b.
[8] Salgarella and Castellan 2020.

machine learning and computer vision, in particular Meta's state-of-the-art interactive segmentation tool, the Segment Anything Model (SAM) (Kirillov et al. 2023), which is pre-trained on a massive and diverse dataset of over 1 billion masks. SAM's generalization and interactivity (allowing user-placed markers that guide the model) might thus allow for semi-automatic annotation of Linear A inscriptions without the need for large amounts of training data, which is unfortunately not available in this context.[9]

[9]By the "interactivity" of SAM, what is meant is NOT any built-in user interface in the form of a GUI, but rather the possibility of prompt-based placement of rectangles of focus, and points of additions and deletions that guide the model's prediction.

1.2 Prior work / Literature review

This thesis places itself at the intersection of computer vision, digital humanities, and epigraphy (the study of inscriptions). Since it is a thesis of engineering, computer vision will remain the primary focus, both in terms of the research questions, and the methods used to answer them. As a result, the other two fields will merely provide the context and motivation for the research. The prior work and initial literature review in focus will thus concentrate on: 1. the relevant theory and state-of-the-art within computer vision (SAM, etc.), 2. the palæographical and digital humanities work that this project will seek to support (SigLA), and 3. the overall context thereof (Linear A).

1.2.1 Research context

For the context of Linear A and its corpus, the background section above can be referred to, citing the sources for: its size (Salgarella 2020) and (Salgarella 2025), its new addition (Del Freo and Zurbach 2025), and where it's accessed (Godart and Olivier 1976–1985). And for the epigraphical and digital humanities work, the background section can also be referred to, citing the sources for SigLA: its paper (Salgarella and Castellan 2020), its about page (SigLA 2021a), and its document repository (SigLA 2021b). Here, additional context can also be provided by Ester Salgarella herself, as she is a co-supervisor of this project, and the main person behind SigLA.

1.2.2 Computer vision literature

As for the relevant theory and state-of-the-art within computer vision, the literature there can be further split into three parts; 1. sources of general image processing, 2. sources concerning SAM and its implementation, and 3. related computer vision script segmentation work for comparison.

For image processing, a general theory book (Gonzalez and Woods 2018) can be cited, as well as more method-specific papers (when other methods are used) like Otsu thresholding (Otsu 1979), and of course the

OpenCV Python documentation (OpenCV Team 2024) for short implementation reference. Then, for SAM, these are: its paper (Kirillov et al. 2023), its lighter implementation MobileSAM (C. Zhang et al. 2023) - and the newest version of that one MobileSAMv2 (Zhao et al. 2023). Lastly, for related segmentation work there exists a paper using deep learning (in contrast to the methodology of this thesis) obtaining results better than state-of-the-art (P. Zhang, Li, and Sun 2024), and a lighter EU-funded historical document segmentation tool *dhSegment* (Oliveira, Seguin, and Kaplan 2018) also using deep learning.

1.3 Problem statement

As previously stated, the aim of this project is to investigate whether parts of the process of creating annotated drawings for SigLA can be automated. More specifically, whether some kind of semi-automatic segmentation tool using interactive and generalized computer vision - specifically the Segment Anything Model (SAM) - can accelerate the workflow, while maintaining high-quality outputs that are still compatible with SigLA's database. But what would be the relative segmentation quality of such a tool? How automatic would it be exactly? And by how much could it improve the efficiency of the workflow? For a decomposition of the problem statement into its three research questions - each accompanied by their respective operationalization - see the following subsection:

1.3.1 Research questions

1. **Segmentation performance**: How well does a SAM-based segmentation approach perform quantitatively in terms of segmentation accuracy, compared to simple classical image processing baselines when applied to Linear A document images? [Operationalization: SAM-based masks will be compared to classical image-processing baselines using overlap metrics (e.g. **IoU!**/**Dice!**) against the ground truth segmentations derived from SigLA's Krita files for a minimum of 25 documents.]

2. **Degree of automation**: To which extent can user interaction be reduced by a SAM-based semi-automatic segmentation workflow, while still maintaining high-quality outputs? [Operationalization: The number of user interactions (e.g. clicks) required to achieve a certain level of segmentation quality (e.g. **IoU!/Dice!**) will be measured against ground truth (SigLA's Krita files) for a minimum of 25 documents.]

3. **Workflow efficiency**: How much reduction in annotation time can be achieved using the proposed tool compared to fully manual digitization, under controlled experimental conditions? [Operationalization: The time it takes Ester Salgarella to annotate manually compared to being assisted by the proposed segmentation tool will be measured for a minimum of 5 documents.]

1.4 Empirical considerations

For the evaluation of the solution, a dataset will be compiled together with Ester Salgarella of at least 25 images of original documents from the corpus, selected and cropped from Cefael. These images are subject to Intellectual Property Rights (IPR) issues, and can thus not be displayed publicly if not through Cefael. However, they can still be used for the development and evaluation of the tool (and can be bought if necessary for show-casing within the thesis). For direct reference to the primary source of data, the specific links to the Cefael online webpages for each document used might have to be cited.

As ground truth, their corresponding digitized layered drawings, stored as Krita (.kra) files on SigLA (provided by Ester Salgarella) will be used and aligned for comparison with tool outputs. The ground truth data might then have to be cleared of any marked "deleted signs" or other annotation elements that our model is not intended to reproduce.

The solution will then be built as a pipeline, consisting of pre-processing, semi-automatic segmentation, post-processing, and export (e.g. as Krita files). At last, the primary deliverable of this project will be delivered to the external partner and co-supervisor Ester Salgarella in the form of a proof-of-concept semi-automatic annotation tool. All supporting components, including evaluation results, and analysis will be documented and presented in the written thesis.

1.5 Impact – innovation and application

The work of this thesis will present itself as a first proof-of-concept for the use of modern computer vision to accelerate the expansion of the SigLA database, supporting the further digitization of the Linear A corpus. Its main contribution will be a semi-automatic annotation tool, which might prove useful to epigraphers for generating segmentations of original document images, as they can then be completed and refined more easily. This approach could potentially also be applied to Linear B, as it is the whole field of Aegean studies that seem to suffer from severe under-digitization of the primary evidence, which is a huge problem for the scientific investigations of scholars. Hopefully, it will inspire further research and development within the field of digital epigraphy, particularly in the application of generalized and interactive computer vision approaches such as SAM on Linear A, and perhaps even in the broader context of use on small corpus ancient writing systems where the lack of big data is ill fit for approaches using deep learning.

CHAPTER 2

Methodology

2.1 Common methodology

2.1.1 Image processing theory

2.1.1.1 Spatial filtering

The basics of spatial filtering can be described by the following equation:

$$g(x, y) = T[f(x, y)]$$

, where $f(x, y)$ (also denoted as $I(x, y)$) is the input image, $g(x, y)$ is the output image.[1]

Convolution is then defined as[2]:

$$(w * f)(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x - s, y - t)$$

, where $w(s, t)$ is the convolution kernel, and a and b are the kernel's half-width and half-height, respectively.

2.1.2 Development environment and tools

Python 3.10 was used for all code development, with the following libraries:

- OpenCV for image processing and computer vision tasks.

[1] Gonzalez and Woods 2018, p. 120.
[2] Gonzalez and Woods 2018, p. 157.

- NumPy for numerical operations and array manipulations.
- Matplotlib for data visualization.
- *Add more!*

2.1.3 Data assumptions (brief reference to Data chapter)

2.1.4 General evaluation metrics

2.1.5 Experimental conventions

2.2 Segmentation quality

The segmentation quality of the tool will be evaluated against a set of baseline models, using a variety of performance metrics. In this section, the configuration of the baseline models, the model to be used by the tool, and the evaluation protocol are described in detail.
[Add some context of RQ1 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

2.2.1 Baseline model configuration

The basline models against which the tool will be evaluated against (from simplest to most advanced) are:

- Otsu's method
- Gaussian adaptive thresholding
- Canny edge detection with morphological filling
- Watershed-based seeded segmentation
- Grabcut-based seeded segmentation

What follows is a brief description of each of these methods, including their characteristis, how they work, and their implementation details.

2.2.1.1 Otsu's method

This is a simple global thresholding method from gray-level histograms originally proposed by Otsu (1979). As the simplest of the baselines, it represents the minimum acceptable segmentation quality, only taking the global pixel intensity into account, with no assumptions about any spatial relationships between pixels. It might therefore perform poorly on images with erosion, uneven lighting, and differences in surface texture, which are the exact characteristics of the Linear A inscriptions to be segmented.[3]

Here is how it works: First, let pixels of the raw image be divided into L shades of gray $[1, 2, ..., L]$. Then, let each pixel be assigned one of two classes C_0 or C_1, guided by a threshold k, where C_0 contains the shades $[1, 2, ..., k]$, and C_1 contains the rest $[k + 1, k + 2, ..., L]$. Otsu's method then works by performing an exhaustive search over all L gray levels to find an optimal threshold k^* that results in a maximum between-class variance σ_B^2:

$$k^* = \underset{1 \leq k < L}{\arg\max}\ \sigma_B^2(k)$$

The between-class variance σ_B^2 can then be defined by a function of the respective total probabilities of the two classes (ω_0 and ω_1) and the mean shade values per class (μ_0 and μ_1):

$$\sigma_B^2(k) = \omega_0(k)\omega_1(k)[\mu_1(k) - \mu_0(k)]^2$$

For the implementation of Otsu's method in OpenCV, used are

- `cv2.THRESH_OTSU` for the thresholding type, and
- `cv2.THRESH_BINARY` for the thresholding method.

[3]Gonzalez and Woods 2018, p. 751.

2.2.1.2 Gaussian adaptive thresholding

For this method, the threshold is not global, but rather computed for each pixel based on the local neighborhood.

$$T(x, y) = (G_\sigma * I)(x, y) - C$$

, where G_σ is the Gaussian kernel, I is the input image, $*$ is the convolution operater, and C is a constant subtracted from the mean to fine-tune the thresholding.

Here,

$$(G_\sigma * I)(x, y) = \sum_{u=-k}^{k} \sum_{v=-k}^{k} G_\sigma(u, v) I(x - u, y - v)$$

, where k is the kernel size, and $G_\sigma(u, v)$ - the value of the Gaussian kernel at position (u, v) - can be defined as:

$$G_\sigma(u, v) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right)$$

For its implementation, used are `cv2.ADAPTIVE_THRESH_GAUSSIAN_C`, and `cv2.THRESH_BINARY` from OpenCV, respectively.

2.2.1.3 Canny edge detection with morphological filling

2.2.1.4 Watershed-based seeded segmentation

2.2.1.5 Grabcut-based seeded segmentation

2.2.2 State-of-the-art model configuration

2.2.2.1 Model architecture

2.2.2.2 Model seeds and preprocessing

2.2.3 Evaluation protocol and performance metrics

$$IoU = \frac{TP}{TP + FP + FN}$$

2.3 Interactive semi-automation

[Add some context of RQ2 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

2.3.1 Interactive tool architecture

2.3.2 Experimental design for automation evaluation

2.3.3 Definition of automation metrics

2.4 Workflow efficiency

[Add some context of RQ3 here, and how the question will be operationalized in bigger picture (though more detail than in introduction).]

2.4.1 Definition of workflow efficiency metrics

2.4.2 Experimental design of the user study

2.4.3 Analysis of results

CHAPTER 3

Data

3.1 Data sources

3.2 Dataset construction

3.3 Ground truth definition

3.4 Data alignment and preprocessing

Raw images were cropped to the bounding box of the inscription, and manually aligned in Krita to overlap as well as possible with the ground truth images.

3.5 Dataset characteristics

3.6 Difficulty stratification

3.7 Dataset limitations and biases

CHAPTER 4

Results

CHAPTER 5

Discussion

CHAPTER 6

Conclusion

6.1 Summary of findings per research question

6.1.1 RQ1: Segmentation performance

6.1.2 RQ2: Degree of automation

6.1.3 RQ3: Workflow efficiency

6.2 Overall contribution

6.3 Final remarks

CHAPTER 7
Future Work

7.1 Methodological extensions

7.2 Dataset expansion and diversification

7.3 Further automation opportunities

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APPENDIX A

An Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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